

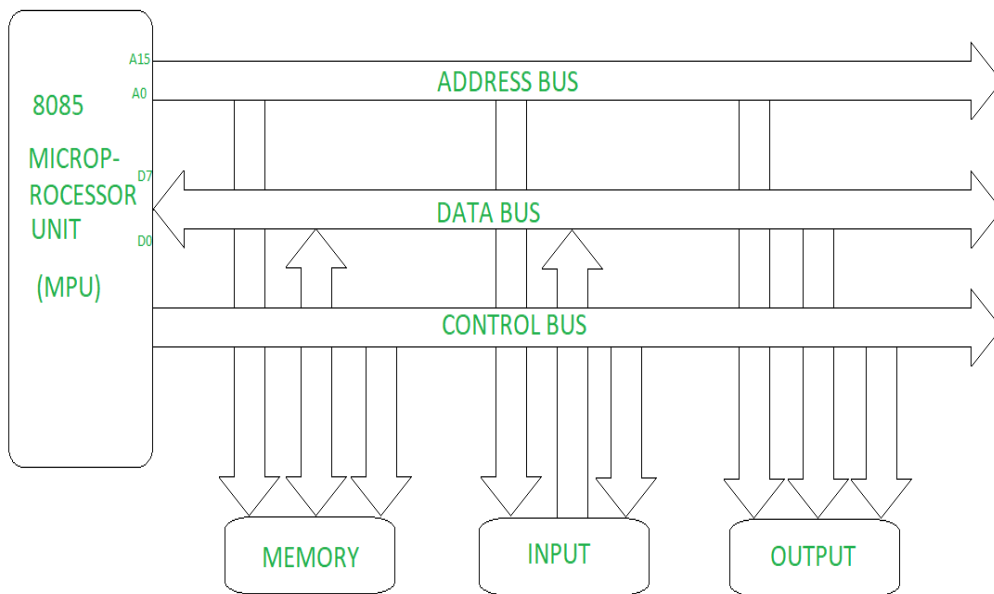
Chapter 3

Bus Structure, Memory and I/O Interfacing

Bus Structure in 8085

Bus is a group of conducting wires which carries information, all the peripherals are connected to microprocessor through Bus.

Diagram to represent bus organization system of 8085 Microprocessor.



Bus organization system of 8085 Microprocessor

There are three types of buses.

1. Address bus –

It is a group of conducting wires which carries address only. Address bus is unidirectional because data flow in one direction, from microprocessor to memory or from microprocessor to Input/output devices (That is, Out of Microprocessor).

Length of Address Bus of 8085 microprocessor is 16 Bit (That is, Four Hexadecimal Digits), ranging from 0000 H to FFFF H, (H denotes Hexadecimal). The microprocessor 8085 can transfer maximum 16 bit address which means it can address 65, 536 different memory location.

The Length of the address bus determines the amount of memory a system can address. Such as a system with a 32-bit address bus can address 2^{32} memory locations. If each memory location holds one byte, the addressable memory space is 4 GB. However, the actual amount of memory that can be accessed is usually much less than this theoretical limit due to chipset and motherboard limitations.

2. Data bus –

It is a group of conducting wires which carries Data only. Data bus is bidirectional because data flow in both directions, from microprocessor to memory or Input/Output devices and from memory or Input/Output devices to microprocessor.

Length of Data Bus of 8085 microprocessor is 8 Bit (That is, two Hexadecimal Digits), ranging from 00 H to FF H. (H denotes Hexadecimal).

When it is write operation, the processor will put the data (to be written) on the data bus, when it is read operation, the memory controller will get the data from specific memory block and put it into the data bus.

The width of the data bus is directly related to the largest number that the bus can carry, such as an 8 bit bus can represent 2 to the power of 8 unique values, this equates to the number 0 to 255. A 16 bit bus can carry 0 to 65535.

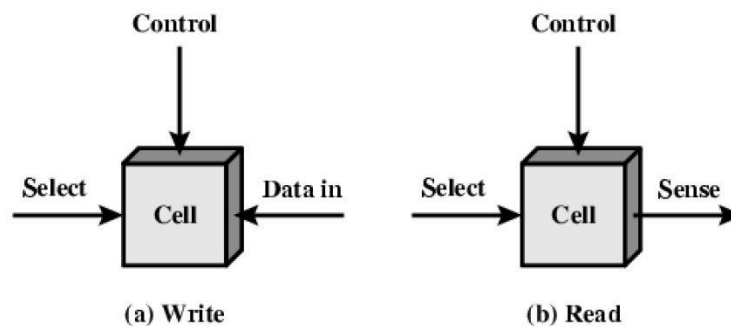
3. Control bus –

It is a group of conducting wires, which is used to generate timing and control signals to control all the associated peripherals, microprocessor uses control bus to process data, that is what to do with selected memory location. Some control signals are:

- Memory read
- Memory write
- I/O read
- I/O Write
- Opcode fetch

Main Memory

- The main memory is the central unit of the computer system. It is relatively large and fast memory to store programs and data during the computer operation. These memories employ semiconductor integrated circuits. The basic element of the semiconductor memory is the memory cell.
- The memory cell has three functional terminals which carries the electrical signal.
 - The select terminal: It selects the cell.
 - The data in terminal: It is used to input data as 0 or 1 and data out or sense terminal is used for the output of the cell's state.
 - The control terminal: It controls the function i.e. it indicates read and write.



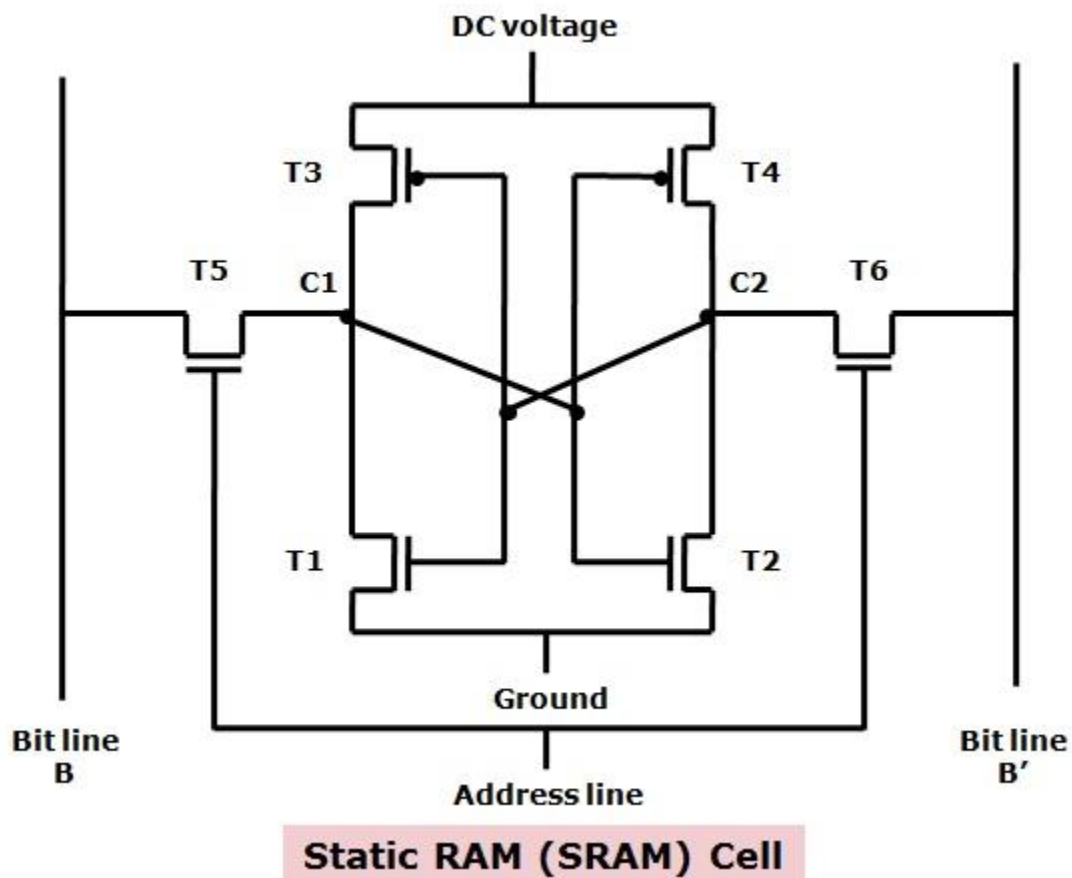
- Most of the main memory in a general purpose computer is made up of RAM integrated circuits chips, but a portion of the memory may be constructed with ROM chips.

RAM– Random Access memory

- Memory cells can be accessed for information transfer from any desired random location.
- The process of locating a word in memory is the same and requires an equal amount of time no matter where the cells are located physically in memory thus named 'Random access'.
- Integrated RAM are available in two possible operating modes, *Static and Dynamic*

Static RAM (SRAM)

- The static RAM consists of flip flop that stores binary information and this stored information remains valid as long as power is applied to the unit.

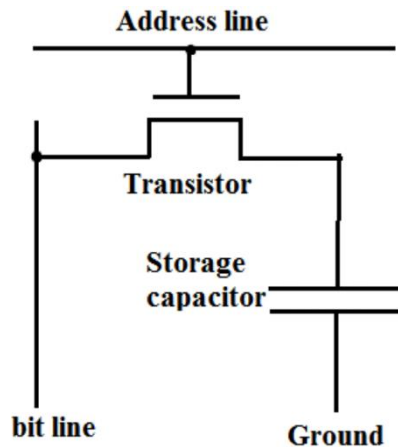


- Four transistors T1, T2, T3 and t4 are cross connected in an arrangement that produces a stable logical state.

- In logic state 1, point C1 is high and point C2 is low. In this state, T1 & T4 are off and T2 & T3 are on.
- In logic state 0, point C1 is low and C2 is high. In this state, T1 & T4 are on and T2 & T3 are off.
- The address line controls the two transistors T5 & T6. When a signal is applied to this line, the two transistors are switched on allowing for read and write operation.
- For a write operation, the desired bit value is applied to line B while it's complement is applied to line B complement. This forces the four transistors T1, T2, T3 & T4 into a proper state.
- For the read operation, the bit value is read from line B.

Dynamic RAM (DRAM)

- The dynamic RAM stores the binary information in the form of electrical charges and capacitor is used for this purpose.
- Since charge stored in capacitor discharges with time, capacitor must be periodically recharged and which is also called refreshing memory.



- The address line is activated when the bit value from this cell is to be read or written.
- The transistor acts as switch that is closed i.e. allowed current to flow, if voltage is applied to the address line; and opened i.e. no current to flow, if no voltage is present in the address line.

For DRAM writing

- The address line is activated which causes the transistor to conduct.
- The sense amplifier senses the content of the data bus line for this cell.
- If the bus line is low, then amplifier will ground the bit line of cell and any charge in capacitor is addressed out.
- If data bus is high, then a +5V is applied on bit line and voltage will flow through transistor and charge the capacitor.

For DRAM reading

- Address line is activated which causes the transistor to conduct.
- If there is charge stored in capacitor, then current will flow through transistor and raise the voltage in bit line. The amplifier will store the voltage and place a 1 on data out line.
- If there is no charge stored in capacitor, then no current will flow through transistor and voltage bit line will not be raised. The amplifier senses that there is no charge and places a 0 on data out line.

SRAM versus DRAM

- Both volatile
Power needed to preserve data
- Static RAM
 - Uses flip flop to store information
 - Needs more space
 - Faster, digital device
 - Expensive, big in size
 - Don't require refreshing circuit
 - Used in cache memory
- Dynamic RAM
 - Uses capacitor to store information
 - More dense i.e. more cells can be accommodated per unit area
 - Slower, analog device
 - Less expensive, small in size
 - Needs refreshing circuit
 - Used in main memory, larger memory units

ROM– Read Only memory

- Read only memory (ROM) contains a permanent pattern of data that cannot be changed.
- A ROM is non-volatile that is no power source is required to maintain the bit values in memory.
- While it is possible to read a ROM, it is not possible to write new data into it.
- The data or program is permanently presented in main memory and never be loaded from a secondary storage device with the advantage of ROM.
- A ROM is created like any other integrated circuit chip, with the data actually wired into the chip as part of the fabrication process.
- It presents two problems
 - The data insertion step includes a relatively large fixed cost, whether one or thousands of copies of a particular ROM are fabricated.
 - There is no room for error. If one bit is wrong, the whole batch of ROM must be thrown out.

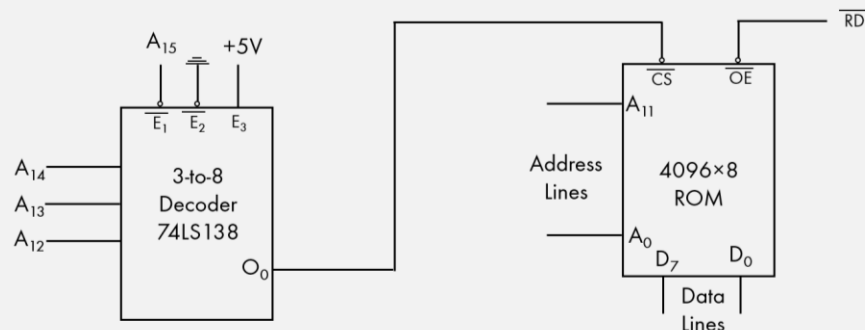
Types of ROM

- Programmable ROM (PROM)**
 It is non-volatile and may be written into only once. The writing process is performed electrically and may be performed by a supplier or customer at a time later than the original chip fabrication.
- Erasable Programmable ROM (EPROM)**
 It is read and written electrically. However, before a write operation, all the storage cells must be erased to the same initial state by exposure of the packaged chip to ultraviolet radiation (UV ray). Erasure is performed by shining an intense ultraviolet light through a window that is designed into the memory chip. EPROM is optically managed and more expensive than PROM, but it has the advantage of the multiple update capability.
- Electrically Erasable programmable ROM (EEPROM)**
 This is a read mostly memory that can be written into at any time without erasing prior contents, only the byte or byte addresses are updated. The write operation takes considerably longer than the read operation, on the order of several hundred microseconds per byte. The EEPROM combines the advantage of non-volatility with the flexibility of being updatable in place, using ordinary bus control, addresses and data lines. EEPROM is more expensive than EPROM and also is less dense, supporting fewer bits per chip.
- Flash Memory**
 Flash memory is also the semiconductor memory and because of the speed with which it can be reprogrammed, it is termed as flash. It is interpreted between EPROM and EEPROM in both cost and functionality. Like EEPROM, flash memory uses an electrical erasing technology. An entire flash memory can be erased in one or a few seconds, which is much faster than EPROM. In addition, it is possible to erase just blocks of memory rather than an entire chip. However, flash memory doesn't provide byte level erasure, a section of memory cells are erased in an action or 'flash'.

Explain how 4 KB ROM is interfaced with 8085 microprocessor.

$$\text{Number of Address line required} = \log_2 4 \times 1024 = 12$$

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	= 0000 H
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	= 0FFF H



Explain how 4 KB ROM and 2 KB RAM are interfaced with 8085 microprocessor starting with address 0000H. (Without foldback memory)

For 4 KB ROM

Number of Address line required = $\log_2 4 \times 1024 = 12$

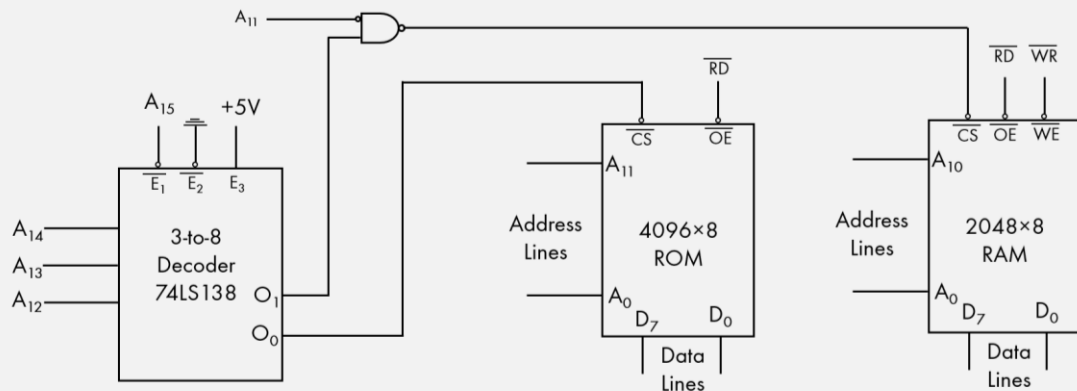
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	= 0000 H
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	= 0FFF H

For 2 KB RAM, assuming A₁₁ is at logic 0.

Number of Address line required = $\log_2 2 \times 1024 = 11$

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	= 1000 H
0	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	

Explain how 4 KB ROM and 2 KB RAM are interfaced with 8085 microprocessor starting with address 0000H. (Without foldback memory)



The interfacing of the I/O devices in 8085 can be done in two ways :

1. Memory-Mapped I/O Interfacing :

In this kind of interfacing, we assign a memory address that can be used in the same manner as we use a normal memory location.

2. I/O Mapped I/O Interfacing :

A kind of interfacing in which we assign an 8-bit address value to the input/output devices which can be accessed using IN and OUT instruction is called I/O Mapped I/O Interfacing.

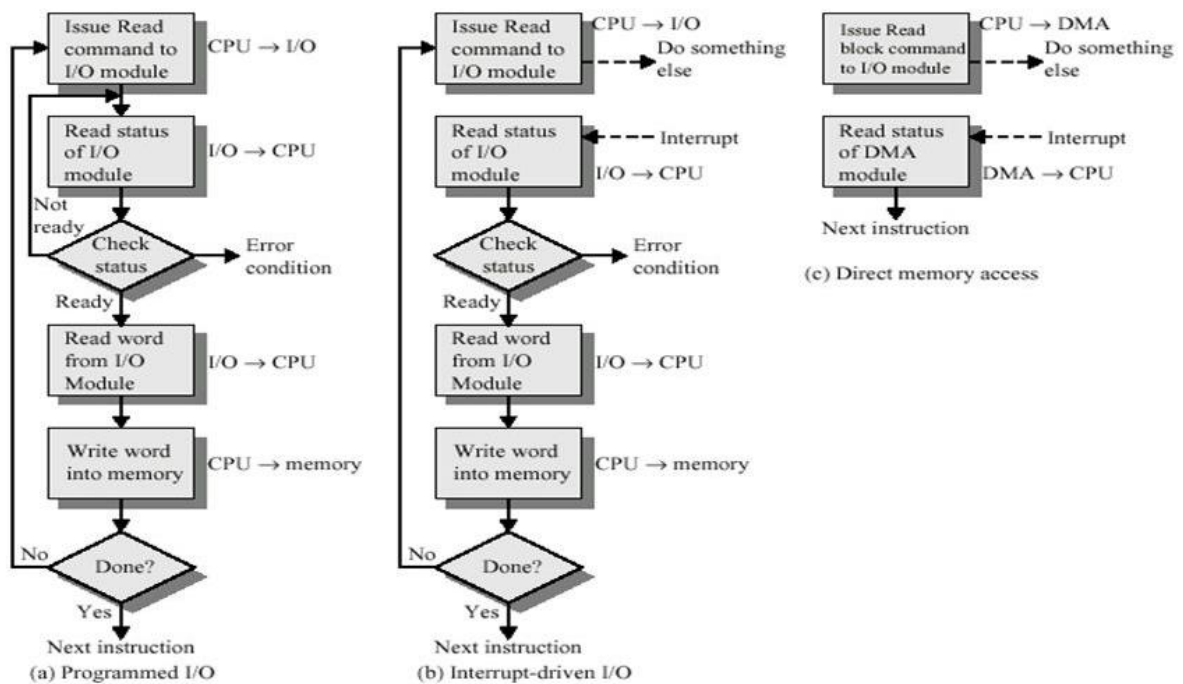
Difference between Memory-Mapped I/O Interfacing and I/O Mapped I/O Interfacing :

Features	Memory Mapped IO	IO Mapped IO
Addressing	IO devices are accessed like any other memory location.	They cannot be accessed like any other memory location.
Address Size	They are assigned with 16-bit address values.	They are assigned with 8-bit address values.
Instructions Used	The instruction used are LDA and STA, etc.	The instruction used are IN and OUT.
Cycles	Cycles involved during operation are Memory Read, Memory Write.	Cycles involved during operation are IO read and IO writes in the case of IO Mapped IO.
Registers Communicating	Any register can communicate with the IO device in case of Memory Mapped IO.	Only Accumulator can communicate with IO devices in case of IO Mapped IO.
Space Involved	2^{16} IO ports are possible to be used for interfacing in case of Memory Mapped IO.	Only 256 I/O ports are available for interfacing in case of IO Mapped IO.
IO/M ['] signal	During writing or read cycles (IO/M ['] = 0) in case of Memory Mapped IO.	During writing or read cycles (IO/M ['] = 1) in case of IO Mapped IO.
Control Signal	No separate control signal required since we have unified memory space in the case of Memory Mapped IO.	Special control signals are used in the case of IO Mapped IO.

Arithmetic and Logical operations	Arithmetic and logical operations are performed directly on the data in the case of Memory Mapped IO.	Arithmetic and logical operations cannot be performed directly on the data in the case of IO Mapped IO.
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Modes of transfer

- Data Transfer between the central computer and I/O devices may be handled in a variety of modes.
- Some modes use CPU as an intermediate path, others transfer the data directly to and from the memory unit.
- Data transfer to and from peripherals may be handled in one of three possible modes.
 - Programmed I/O
 - Interrupt Driven I/O
 - Direct Memory Access (DMA)



Programmed I/O

- Programmed I/O operations are the result of I/O instructions written in the computer program.
- In programmed I/O, each data transfer is initiated by the instructions in the CPU and hence the CPU is in the continuous monitoring of the interface.

- Input instruction is used to transfer data from I/O device to CPU, store instruction is used to transfer data from CPU to memory and output instruction is used to transfer data from CPU to I/O device.
- This technique is generally used in very slow speed computer and is not a efficient method if the speed of the CPU and I/O is different.

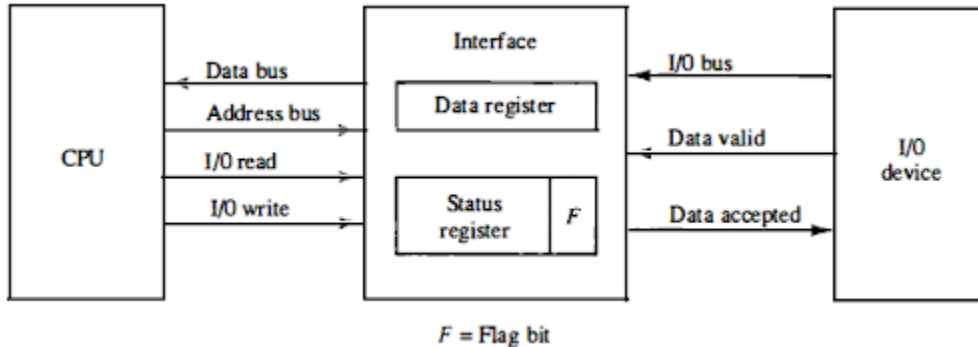


Fig: Data transfer from I/O device to CPU

- I/O device places the data on the I/O bus and enables its data valid signal
- The interface accepts the data in the data register and sets the F bit of status register and also enables the data accepted signal.
- Data valid line is disabled by I/O device.
- CPU is in a continuous monitoring of the interface in which it checks the F bit of the status register.
 - If it is set i.e. 1, then the CPU reads the data from data register and sets F bit to zero
 - If it is reset i.e. 0, then the CPU remains monitoring the interface.
- Interface disables the data accepted signal and the system goes to initial state where next item of data is placed on the data bus.

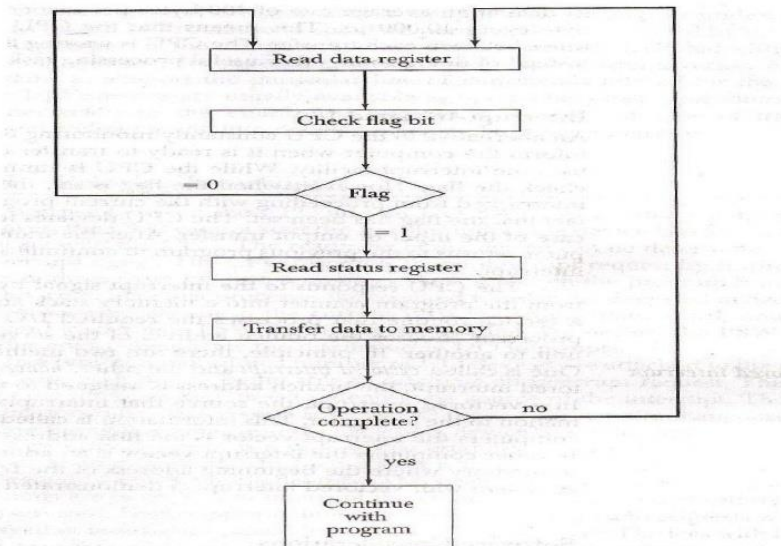


Fig: flowchart for CPU program to input data

Characteristics:

- Continuous CPU involvement
- CPU slowed down to I/O speed
- Simple

- Least hardware

Polling, or polled operation, in computer science, refers to actively sampling the status of an external device by a client program as a synchronous activity. Polling is most often used in terms of input/output (I/O), and is also referred to as **polled I/O or software driven I/O**.

Interrupt-driven I/O

- Polling takes valuable CPU time
- Open communication only when some data has to be passed -> *Interrupt*.
- I/O interface, instead of the CPU, monitors the I/O device
- When the interface determines that the I/O device is ready for data transfer, it generates an *Interrupt Request* to the CPU
- Upon detecting an interrupt, CPU stops momentarily the task it is doing, branches to the service routine to process the data transfer, and then returns to the task it was performing

The problem with programmed I/O is that the processor has to wait a long time for the I/O module of concern to be ready for either reception or transmission of data. The processor, while waiting, must repeatedly interrogate the status of the I/O module. As a result, the level of the performance of the entire system is severely degraded. An alternative is for the processor to issue an I/O command to a module and then go on to do some other useful work. The I/O module will then interrupt the processor to request service when it is ready to exchange data with processor. The processor then executes the data transfer, and then resumes its former processing. The interrupt can be initiated either by software or by hardware.

Interrupt Driven I/O basic operation

- CPU issues read command
- I/O module gets data from peripheral whilst CPU does other work
- I/O module interrupts CPU
- CPU requests data
- I/O module transfers data

Interrupt Processing from CPU viewpoint

- Issue read command
- Do other work
- Check for interrupt at end of each instruction cycle
- If interrupted:-
 - Save context (registers)
 - Process interrupt
 - Fetch data & store

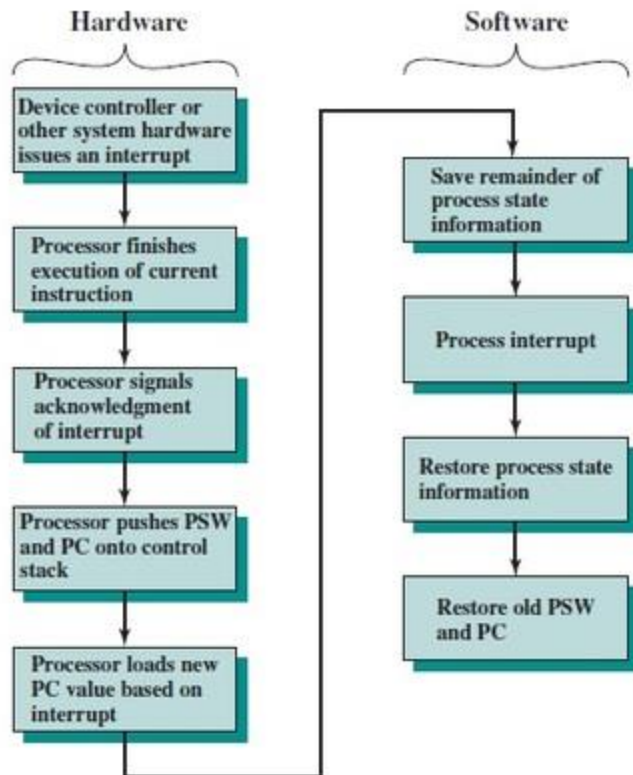


Fig: Simple Interrupt Processing

8259 Programmable Interrupt Controller

There are 5 hardware interrupts and 2 hardware interrupts in 8085 and 8086 respectively. But by connecting 8259 with CPU, we can increase the interrupt handling capability. 8259 combines the multi interrupt input sources into a single interrupt output. Interfacing of single PIC provides 8 interrupts inputs from IR0-IR7.

For example, interfacing of 8085 and 8259 increases the interrupt handling capability of 8085 microprocessor from 5 to 8 interrupt levels.

Features of 8259 PIC microprocessor –

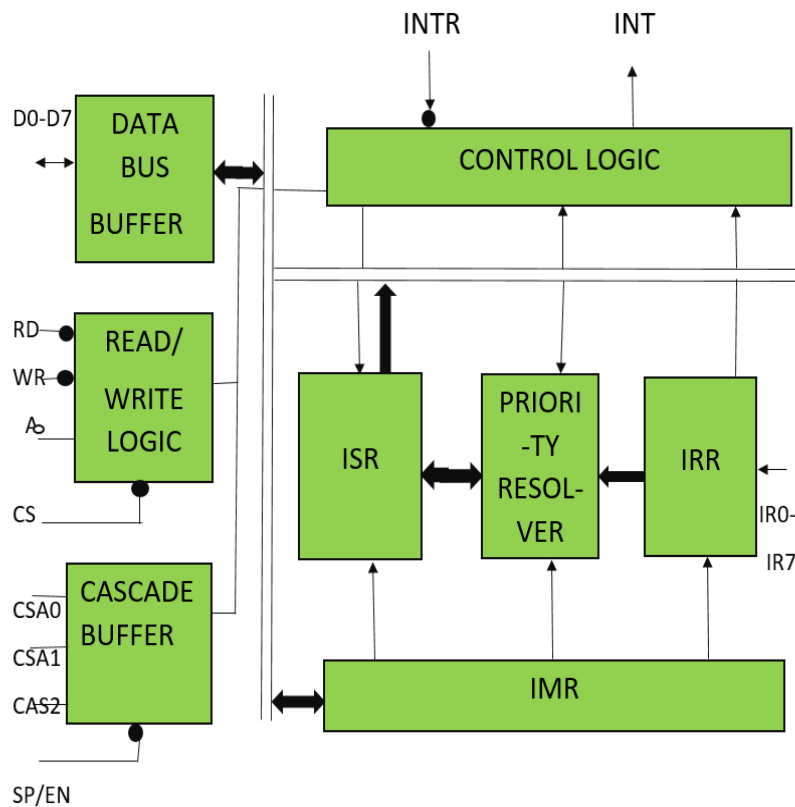
1. Intel 8259 is designed for Intel 8085 and Intel 8086 microprocessor.
2. It can be programmed either in level triggered or in edge triggered interrupt level.
3. We can mask individual bits of interrupt request register.
4. We can increase interrupt handling capability upto 64 interrupt level by cascading further 8259 PIC.
5. Clock cycle is not required.

Pin Diagram of 8259 –

$\overline{CS}$	1	8259 PIC	28	V_{CC}
$\overline{WR}$	2		27	A_0
$\overline{RD}$	3		26	$\overline{INTA}$
D7	4		25	IR7
D6	5		24	IR6
D5	6		23	IR5
D4	7		22	IR4
D3	8		21	IR3
D2	9		20	IR2
D1	10		19	IR1
D0	11		18	IR0
CAS0	12		17	INT
CAS1	13		16	$\overline{SP/EN}$
Gnd	14		15	CAS2

We can see through above diagram that there are total 28 pins in 8259 PIC microprocessor where V_{CC} :5V Power supply and Gnd: ground. Other pins use are explained below.

Block Diagram of 8259 PIC microprocessor –



The Block Diagram consists of 8 blocks which are – Data Bus Buffer, Read/Write Logic, Cascade Buffer Comparator, Control Logic, Priority Resolver and 3 registers- ISR, IRR, IMR.

1. **Data bus buffer**

This Block is used as a mediator between 8259 and 8085/8086 microprocessor by acting as a buffer. It takes the control word from the 8085 (let say) microprocessor and transfer it to the control logic of 8259 microprocessor. Also, after selection of Interrupt by 8259 microprocessor, it transfer the opcode of the selected Interrupt and address of the Interrupt service sub routine to the other connected microprocessor. The data bus buffer consists of 8 bits represented as D0-D7 in the block diagram. Thus, shows that a maximum of 8 bits data can be transferred at a time.

2. **Read/Write logic**

This block works only when the value of pin CS is low (as this pin is active low). This block is responsible for the flow of data depending upon the inputs of RD and WR. These two pins are active low pins used for read and write operations.

3. **Control logic**

It is the centre of the microprocessor and controls the functioning of every block. It has pin INTR which is connected with other microprocessor for taking interrupt request and pin INT for giving the output. If 8259 is enabled, and the other microprocessor Interrupt flag is high then this causes the value of the output INT pin high and in this way 8259 responds to the request made by other microprocessor.

4. **Interrupt request register (IRR)**

It stores all the interrupt level which are requesting for Interrupt services.

5. **Interrupt service register (ISR)**

It stores the interrupt level which are currently being executed.

6. **Interrupt mask register (IMR)**

It stores the interrupt level which have to be masked by storing the masking bits of the interrupt level.

7. **Priority resolver**

It examines all the three registers and set the priority of interrupts and according to the priority of the interrupts, interrupt with highest priority is set in ISR register. Also, it reset the interrupt level which is already been serviced in IRR.

8. **Cascade buffer**

To increase the Interrupt handling capability, we can further cascade more number of pins by using cascade buffer. So, during increment of interrupt capability, CSA lines are used to control multiple interrupt structure.

SP/EN (Slave program/Enable buffer) pin is when set to high, works in master mode else in slave mode. In Non Buffered mode, SP/EN pin is used to specify whether 8259 work as master or slave and in Buffered mode, SP/EN pin is used as an output to enable data bus.

Direct Memory access

- Large blocks of data transferred at a high speed to or from high speed devices, magnetic drums, disks, tapes, etc.
- DMA controller Interface that provides I/O transfer of data directly to and from the memory and the I/O device
- CPU initializes the DMA controller by sending a memory address and the number of words to be transferred

- Actual transfer of data is done directly between the device and memory through DMA controller
-> Freeing CPU for other tasks

The transfer of data between the peripheral and memory without the interaction of CPU and letting the peripheral device manage the memory bus directly is termed as Direct Memory Access (DMA).

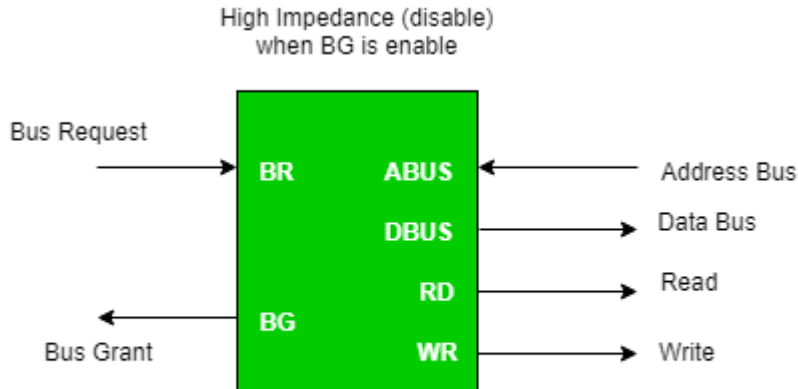


Figure - CPU Bus Signals for DMA Transfer

The two control signals Bus Request and Bus Grant are used to facilitate the DMA transfer. The bus request input is used by the DMA controller to request the CPU for the control of the buses. When BR signal is high, the CPU terminates the execution of the current instructions and then places the address, data, read and write lines to the high impedance state and sends the bus grant signal. The DMA controller now takes the control of the buses and transfers the data directly between memory and I/O without processor interaction. When the transfer is completed, the bus request signal is made low by DMA. In response to which CPU disables the bus grant and again CPU takes the control of address, data, read and write lines.

The 8237 DMA Controller

- The 8237 supplies memory & I/O with control signals and memory address information during the DMA transfer.
- It is actually a special-purpose microprocessor whose job is high-speed data transfer between memory and I/O
- 8237 is not a discrete component in modern microprocessor-based systems.
- It appears within many system controller chip sets
- 8237 is a four-channel device compatible with 8086/8088, adequate for small systems.
- Expandable to any number of DMA channel inputs
- 8237 is capable of DMA transfers at rates up to 1.6MB per second.
- Each channel is capable of addressing a full 64K-byte section of memory.

8255A – Programmable Peripheral Interface

The 8255A is a general purpose programmable I/O device designed to transfer the data from I/O to interrupt I/O under certain conditions as required. It can be used with almost any microprocessor.

It consists of three 8-bit bidirectional I/O ports (24I/O lines) which can be configured as per the requirement.

Ports of 8255A

8255A has three ports, i.e., PORT A, PORT B, and PORT C.

- **Port A** contains one 8-bit output latch/buffer and one 8-bit input buffer.
- **Port B** is similar to PORT A.
- **Port C** can be split into two parts, i.e. PORT C lower (PC0-PC3) and PORT C upper (PC7-PC4) by the control word.

These three ports are further divided into two groups, i.e. Group A includes PORT A and upper PORT C. Group B includes PORT B and lower PORT C. These two groups can be programmed in three different modes, i.e. the first mode is named as mode 0, the second mode is named as Mode 1 and the third mode is named as Mode 2.

Operating Modes

8255A has three different operating modes –

- **Mode 0** – In this mode, Port A and B is used as two 8-bit ports and Port C as two 4-bit ports. Each port can be programmed in either input mode or output mode where outputs are latched and inputs are not latched. Ports do not have interrupt capability.
- **Mode 1** – In this mode, Port A and B is used as 8-bit I/O ports. They can be configured as either input or output ports. Each port uses three lines from port C as handshake signals. Inputs and outputs are latched.
- **Mode 2** – In this mode, Port A can be configured as the bidirectional port and Port B either in Mode 0 or Mode 1. Port A uses five signals from Port C as handshake signals for data transfer. The remaining three signals from Port C can be used either as simple I/O or as handshake for port B.

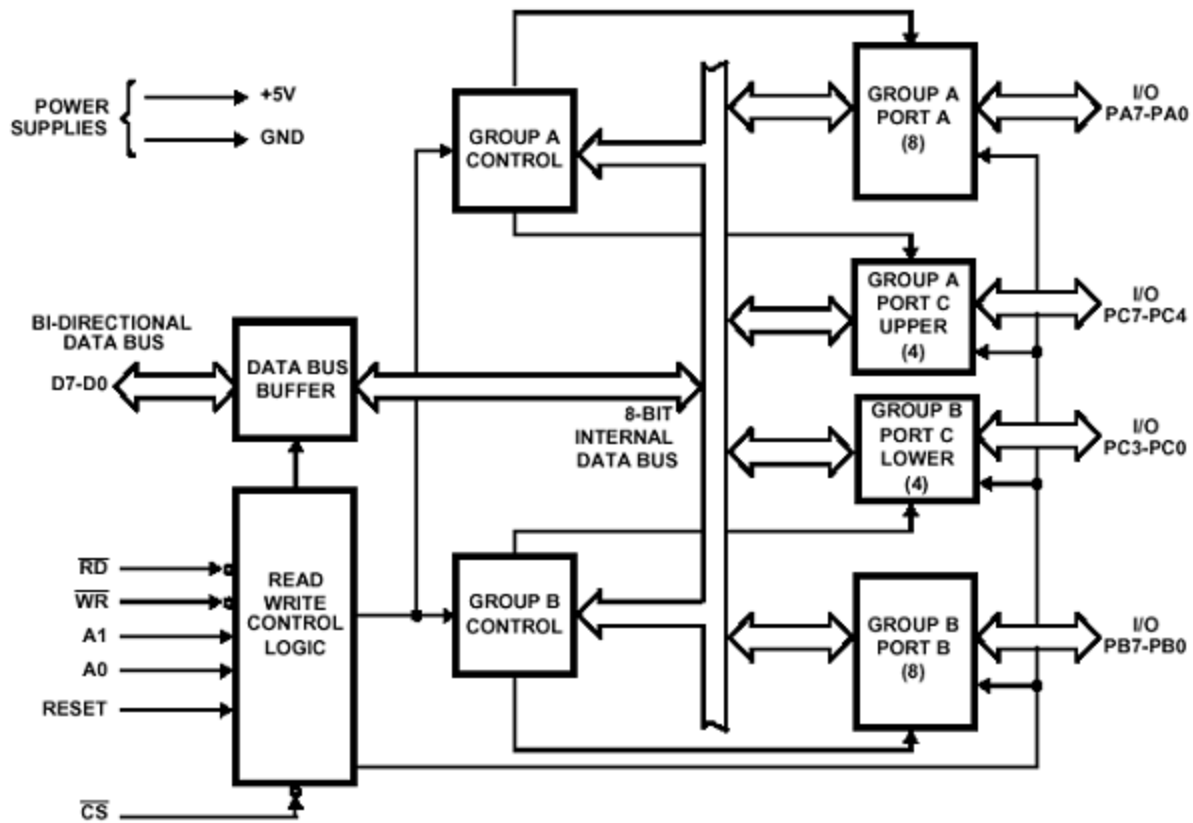
Features of 8255A

The prominent features of 8255A are as follows –

- It consists of 3 8-bit IO ports i.e. PA, PB, and PC.
- Address/data bus must be externally demuxed.
- It is TTL compatible.
- It has improved DC driving capability.

8255 Internal Architecture

The following figure shows the architecture of 8255A.



Data Bus Buffer

- This tri-state bidirectional 8-bit buffer is used to interface 8255A to the system data bus.
- Data, control words and status information are transferred through data bus buffer.

Read/Write Control Logic

- The function of this block is to manage all the internal and external transfer of both data and control words.
- It has six lines:

RD(Read): When the signal is low, the MPU reads data from the selected I/O port of 8255A.

WR(Write): When the signal is low, the MPU writes data into the selected I/O port or writes control word into control register.

RESET (Reset): This is an active high signal. It cleans the control register and set all ports in the input mode.

CS(Chip Select), **A₀** and **A₁**: These are the device select signals. CS is connected to decoded address; A₀ and A₁ are generally connected to MPU address lines A₀ and A₁ respectively.

If CS = 0, 8255A is selected and if CS = 1, 8255A is not selected.

A₀ and A₁ specify the selection of I/O ports as shown below.

A ₁	A ₀	Port Selected
0	0	Port A
0	1	Port B
1	0	Port C
1	1	Control Register

Group A and Group B Controls

- Each of the control blocks accepts commands from read/write control logic and issues proper command to its associated ports.

Group A control: Port A and Port C upper

Group B control: Port B and Port C lower

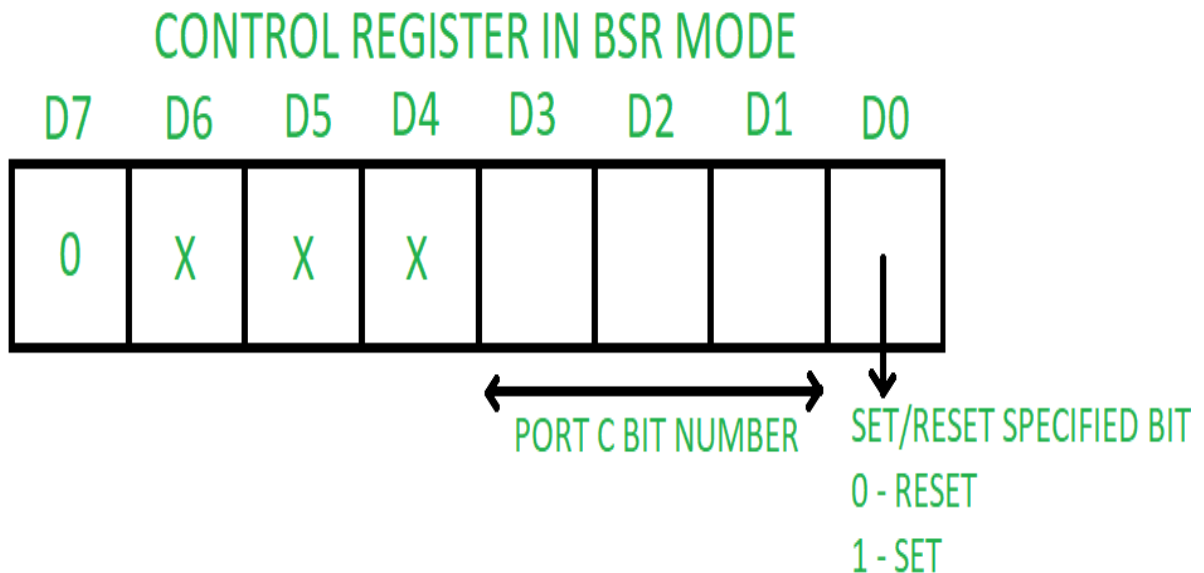
I/O ports

- The 8255A has 24 I/O pins that are grouped into 8-bit parallel I/O ports : Port A(PA) Port B(PB) and Port C(PC)
- The 8-bit of port C can be used as individual bits or grouped into two 4-ports: Port C upper(PC_U) and Port C lower(PC_L)
- The functions of I/O ports are defined by writing a control word into control register.

Operating modes

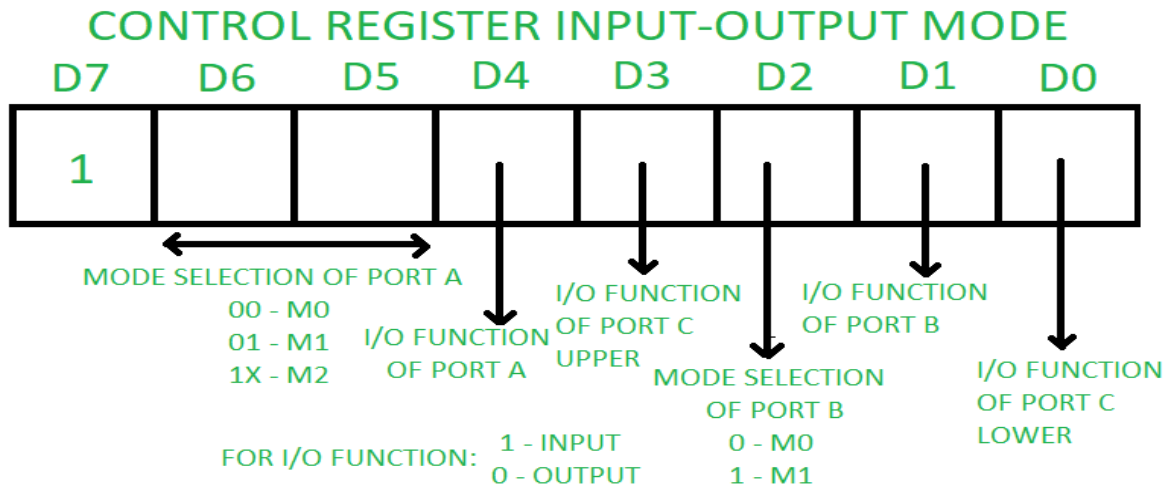
1. Bit set reset (BSR) mode

If MSB of control word (D7) is 0, PPI works in BSR mode. In this mode only port C bits are used for set or reset.



2. Input-Output mode

If MSB of control word (D7) is 1, PPI works in input-output mode. This is further divided into three modes:



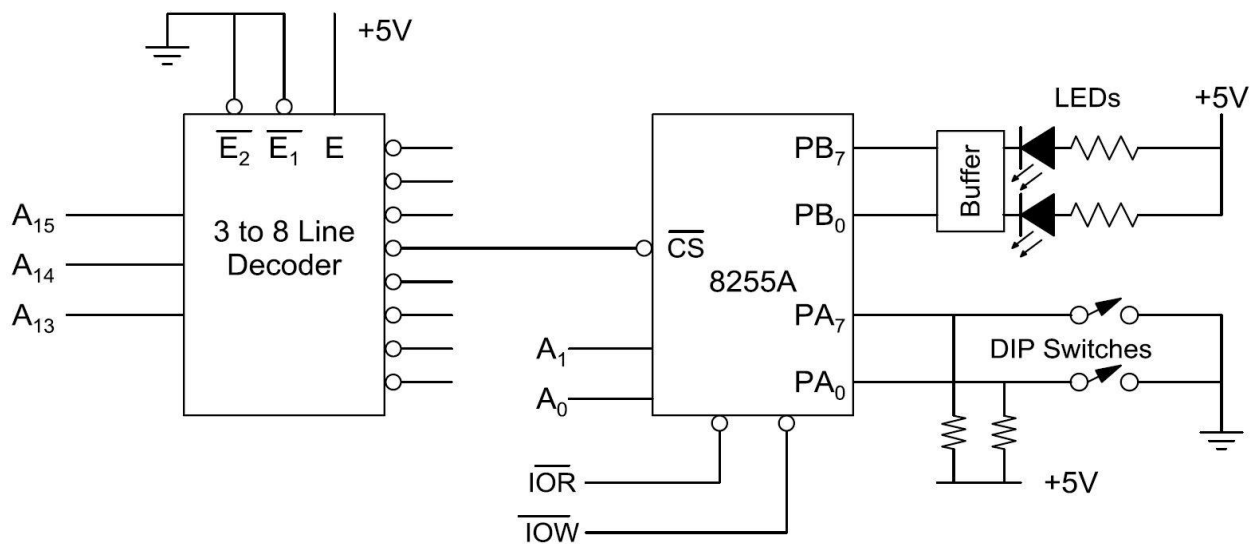
- **Mode 0** – In this mode all the three ports (port A, B, C) can work as simple input function or simple output function. In this mode there is no interrupt handling capacity.
- **Mode 1** – Handshake I/O mode or strobed I/O mode. In this mode either port A or port B can work as simple input port or simple output port, and port C bits are used for handshake signals before actual data transmission. It has interrupt handling capacity and input and output are latched.

Example: A CPU wants to transfer data to a printer. In this case since speed of processor is very fast as compared to relatively slow printer, so before actual data transfer it will send handshake signals to the printer for synchronization of the speed of the CPU and the peripherals.



- **Mode 2** – Bi-directional data bus mode. In this mode only port A works, and port B can work either in mode 0 or mode 1. 6 bits port C are used as handshake signals. It also has interrupt handling capacity.

- Derive a control word for 8255A to configure:
 Port A= i/p port
 Port B= o/p port
 Port C Lower= i/p port
 Port C Upper= o/p port
 In simple I/O mode with Group A in mode 0 and Group B in mode 1.
 Solution:
 Control Word =
 =10010101
 = 95H
- Identify the port addresses in given figure.
 - Identify the Mode 0 control word to configure port A as an input port and port B as an output port.
 - Write a program to read the Dip switches and display the reading from port A at port B.



Solution

a) This is I/O mapped I/O; when A15 A14 A13 is 011, then chip select of 8255 is enabled. We also know that during the execution of IN and OUT instruction, A15-A8 and AD7-AD0 carry the same signals. Keeping this in mind, port addresses will be derived. Firstly, port A's port address will be calculated as under:

A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
0	1	1	X	X	X	X	X	X	X	X	X	X	X	0	0

To have equality, 0's and 1's on one side of the equation must appear on other sides. This means that AD7 AD6 AD5 must equal 011 and A9 and A8 must equal 00 (port A) to get

0 1 1 X X X 00 = 0 1 1 X X X 00

Since the remaining don't cares can be 0's and 1's, there are many solutions. For instance, if all the don't cares are equal to zero; address of port A becomes 1110 0000 (60H). The port addresses of the given figure are determined as under:

Port A = 60H

Port B = 61H

Port C = 62H

Control Register = 63H

b) The Mode 0 control word to configure port A input and port B output is calculated as under:

= 90H

c) Program subroutine to read DIP switches and display the reading from port A at port B is as under:

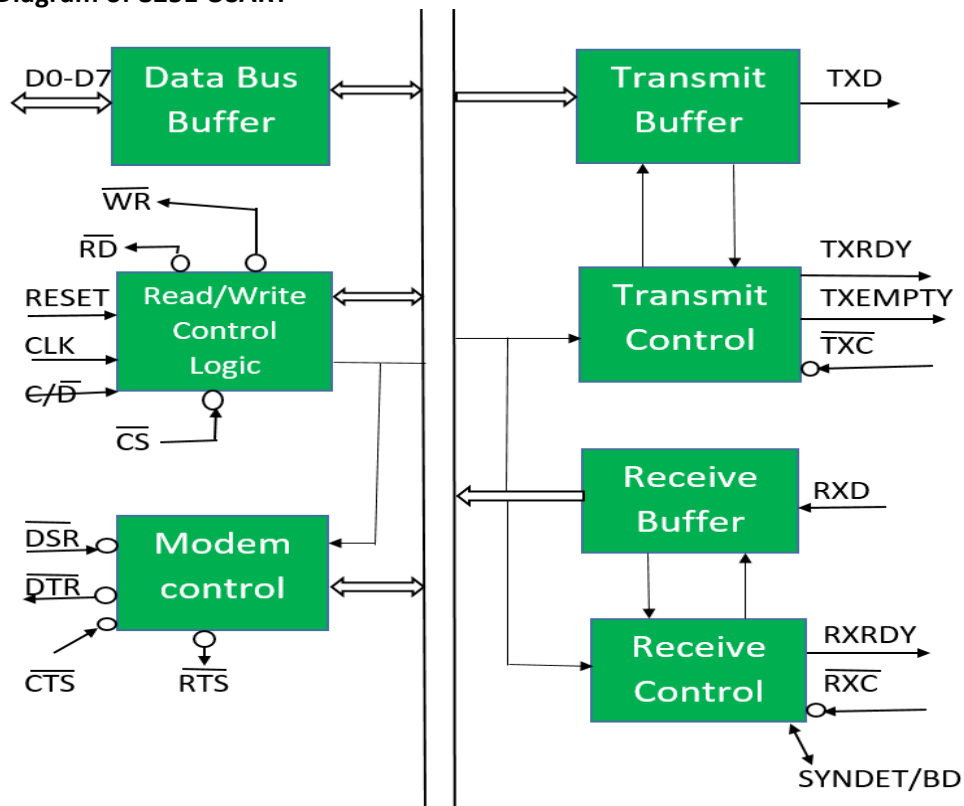
```
MVI A, 90H;    Load ACC with the control word
OUT 63H;       Write the control word in control register and initialize the ports
IN 60H;        Read switches at port A
OUT 61H;       Display the reading at port B
RET
```

8251 USART

8251 universal synchronous asynchronous receiver transmitter (USART) acts as a mediator between microprocessor and peripheral to transmit serial data into parallel form and vice versa.

1. It takes data serially from peripheral (outside devices) and converts into parallel data.
2. After converting the data into parallel form, it transmits it to the CPU.
3. Similarly, it receives parallel data from microprocessor and converts it into serial form.
4. After converting data into serial form, it transmits it to outside device (peripheral).

Block Diagram of 8251 USART –



It contains the following blocks:

1. **Data bus buffer –**

This block helps in interfacing the internal data bus of 8251 to the system data bus. The data transmission is possible between 8251 and CPU by the data bus buffer block.

2. **Read/Write control logic –**

It is a control block for overall device. It controls the overall working by selecting the operation to be done. The operation selection depends upon input signals as:

$\overline{CS}$	$C/\overline{D}$	$\overline{RD}$	$\overline{WR}$	Operation
1	X	X	X	Invalid
0	0	0	1	data CPU < ----- 8251
0	0	1	0	data CPU ----- > 8251
0	1	0	1	Status word CPU < -----8251
0	1	1	0	Control word CPU----- > 8251

In this way, this unit selects one of the three registers- data buffer register, control register, status register.

3. **Modem control (modulator/demodulator) –**

A device converts analog signals to digital signals and vice-versa and helps the computers to communicate over telephone lines or cable wires. The following are active-low pins of Modem.

- **DSR:** Data Set Ready signal is an input signal.
- **DTR:** Data terminal Ready is an output signal.
- **CTS:** It is an input signal which controls the data transmit circuit.
- **RTS:** It is an output signal which is used to set the status RTS.

4. **Transmit buffer**

This block is used for parallel to serial converter that receives a parallel byte for conversion into serial signal and further transmission onto the common channel.

- **TXD:** It is an output signal, if its value is one, means transmitter will transmit the data.

5. **Transmit control**

This block is used to control the data transmission with the help of following pins:

- **TXRDY:** It means transmitter is ready to transmit data character.
- **TXEMPTY:** An output signal which indicates that TXEMPTY pin has transmitted all the data characters and transmitter is empty now.
- **TXC:** An active-low input pin which controls the data transmission rate of transmitted data.

6. **Receive buffer**

This block acts as a buffer for the received data.

- **RXD:** An input signal which receives the data.

7. **Receive control**

This block controls the receiving data.

- **RXRDY:** An input signal indicates that it is ready to receive the data.
- **RXC:** An active-low input signal which controls the data transmission rate of received data.

- **SYNDET/BD:** An input or output terminal. External synchronous mode-input terminal and asynchronous mode-output terminal.